

HTML5 Exercises with Solutions

Exercise 1: Create a Basic Web Page

Task

Create a webpage with:

- A title
- A main heading
- A paragraph

Exercise 2: Working with Lists

Task

Create:

- An unordered list of 3 fruits
- An ordered list of 3 programming languages

Exercise 3: Create a Table

Task

Create a table displaying:

- Name
- Age
- City

Exercise 4: Add Images and Links

Task

Create:

- An image
- A hyperlink to Google

Exercise 5: Student Registration Form

Task

Create a form with:

- Name
- Email
- Password
- Submit button

Exercise 6: Use Semantic Elements

Task

Create a webpage using:

- header
- nav
- main
- article
- footer

Exercise 7: Audio and Video

Task

Add:

- One audio file
- One video file

Exercise 8: Create a Survey Form

Task

Use:

- Radio buttons
- Checkboxes

- Textarea

Exercise 9: Build a Timetable

Task

Create a timetable for three days and three periods.

Exercise 10: Portfolio Website

Task

Create a portfolio page containing:

- About section
- Skills section
- Projects section
- Contact form

HTML5 Analytical Thinking Exercises (HTML Only)

These exercises focus on **problem-solving, document structure, accessibility, semantics, and choosing the correct HTML5 elements**, rather than simply writing tags.

Exercise 1: Design a University Website Structure

Scenario

A university website contains:

- University name and logo
- Main navigation
- Latest announcements
- Upcoming events
- Student testimonials
- Copyright information

Task

Create a semantic HTML5 structure using the most appropriate elements.

Analytical Challenge

Decide:

- Which content belongs inside <header>?
 - Should announcements and events be <section> or <article>?
 - Where should testimonials be placed?
-

Exercise 2: News Portal Information Architecture

Scenario

A news portal displays:

- Breaking news
- Politics news
- Sports news
- Technology news
- Advertisements
- Related articles

Task

Create the semantic HTML structure.

Analytical Challenge

Determine:

- Which content belongs inside <main>?

- Which content should be placed inside <aside>?
 - Which items should be represented by <article>?
-

Exercise 3: E-Commerce Product Page

Scenario

A product page contains:

- Product title
- Product images
- Product specifications
- Customer reviews
- Related products

Task

Build the page structure using HTML5 semantic elements.

Analytical Challenge

Justify:

- Why specifications should or should not be a table.
 - Whether reviews are separate articles.
 - Whether related products belong in <aside>.
-

Exercise 4: Conference Registration Form

Scenario

A conference requires:

- Personal details
- Professional details
- Session selection
- Dietary preferences
- Emergency contact

Task

Design the form.

Analytical Challenge

Choose:

- Appropriate input types.
- Which fields should be grouped using <fieldset>.
- Which fields require validation attributes.

Exercise 5: Blog Homepage Content Modeling

Scenario

A blog homepage contains:

- Site title
- Navigation menu
- Featured article
- Recent articles
- Popular tags
- Newsletter subscription form

Task

Create a semantic page structure.

Analytical Challenge

Identify:

- Which content is primary.

- Which content is supplementary.
- Where <aside> should be used.

Exercise 6: Employee Directory

Scenario

Create an employee directory showing:

- Employee ID
- Name
- Department
- Email
- Phone number

Task

Represent the data appropriately.

Analytical Challenge

Determine:

- Why a table is preferable to lists.
 - How to use <thead>, <tbody>, and <caption> effectively.
-

Exercise 7: Online Examination Form

Scenario

An online exam contains:

- Multiple-choice questions
- True/False questions
- Short-answer questions

Task

Create the form structure.

Analytical Challenge

Decide:

- When to use radio buttons.
 - When to use textareas.
 - How to group questions semantically.
-

Exercise 8: Restaurant Reservation System

Scenario

Customers can:

- Select date
- Select time
- Enter number of guests
- Provide special requests

Task

Create a reservation form.

Analytical Challenge

Select:

- Appropriate HTML5 input types.
 - Required fields.
 - Constraints for guest count.
-

Exercise 9: Accessibility-Focused Job Application Form

Scenario

Build a job application form for users who may rely on screen readers.

Task

Create an accessible HTML-only solution.

Analytical Challenge

Ensure:

- Every field has a label.
 - Related controls are grouped.
 - Form instructions are clear.
 - Semantic structure improves accessibility.
-

Exercise 10: Digital Magazine Layout

Scenario

A digital magazine contains:

- Cover story
- Feature stories
- Editor's note
- Sidebar recommendations
- Reader comments

Task

Design the page structure.

Analytical Challenge

Decide:

- Which content should be articles.
 - Which content belongs in an aside.
 - How comments should be organized semantically.
-

Bonus Challenge (Advanced)

Exercise 11: Complete Learning Management System (LMS)

Scenario

Create the HTML structure for a learning platform containing:

- Header
- Navigation
- Dashboard
- Courses
- Lessons
- Assignments
- Announcements
- Student Profile
- Footer

Constraints

Use only:

- Semantic HTML5 elements
- Tables where appropriate
- Forms where appropriate
- Lists where appropriate

Analytical Questions

1. Which areas belong inside <main>?
2. Should lessons be represented as <article> elements?
3. Which information belongs in <aside>?

4. How should assignments be organized semantically?
 5. How can accessibility be improved without CSS or JavaScript?
-

Sample Solution (Exercise 1)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>University Website</title>
</head>
<body>

<header>
  <h1>ABC University</h1>

  <nav>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">Admissions</a></li>
      <li><a href="#">Departments</a></li>
    </ul>
  </nav>
</header>

<main>

  <section>
    <h2>Latest Announcements</h2>

    <article>
      <h3>Semester Registration Open</h3>
      <p>Students may register until July 15.</p>
    </article>

    <article>
      <h3>Scholarship Applications</h3>
      <p>Applications are now available.</p>
    </article>
  </section>

  <section>
    <h2>Upcoming Events</h2>

    <article>
      <h3>Research Symposium</h3>
      <p>July 20, 2026</p>
    </article>
  </section>
```

```
<section>
  <h2>Student Testimonials</h2>

  <article>
    <blockquote>
      Excellent learning environment.
    </blockquote>
    <p>— Student A</p>
  </article>
</section>
```

```
</main>
```

```
<footer>
  <p>&copy; 2026 ABC University</p>
</footer>
```

```
</body>
</html>
```

These exercises are suitable for **interviews, lab exams, university assessments, and HTML5 semantic design evaluations** because they require students to analyze content structure before writing HTML.